INVESTOR RELATIONS

January 2026

Cellumed

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Chapter 1. Company Overview

Overview

Company	CELLUMED Co., Ltd.
CEO	YOU IN SOO
Foundation	AUG 1st, 1997
Capital	US$ 29 Million
Main Products	Dental Powder, CELLUDERM Series Rafugen DBM, BioBMP2, TKR
Employee	83
Location	Head Quarter: Seoul, Korea GMP Factory: Seoul, Korea
Web Site	www.cellumed.co.kr

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History

2020's

2023 Developed CELLUDERM series

2022 Research for the production of enzymes essential for the synthesis of mRNA-based vaccines and therapeutics

2021 Dentgen, RenewOss, and RegenOss approved by the Ministry of Food and Drug Safety

2010's

2019 BIOFLEXGIDE & BIOFIX Composite Screw Approved by the Ministry of Food and Drug Safety

2018 Rafugen™ DBM US FDA Approval

BMG (Bio Meniscus Graft) approved by the Ministry of Food and Drug Safety

2017 Launch of Solid W6 (DBM Wedge Block)

Launch of Bio Absorbable Screw

2016 Viscoseal Syringe Medical Device Approval

LF Fixed Knee (TKR) FDA Application

2015 CHO.rhBMP2 approved by the Thai FDA

Launch of LF Fixed Knee

2010 Approval for manufacturing artificial knee joints

CHO.rhBMP-2 (NO.13-922) manufacturing approval

2000's

2009 KOSDApproval of Human Tissue Bank (No. 11) from the Ministry of Food and Drug Safety / Approval of Imported Human Tissue BankAQ listing

Acquisition of ENDOTEC, an artificial knee joint company

2007 Medical device manufacturing license

2006 Establishment of an R&D laboratory

2005 Approval of Human Tissue Bank (No. 11) from the Ministry of Food and Drug Safety / Approval of Imported Human Tissue Bank

1990's

1997 Establishment of Korea Bone Bank

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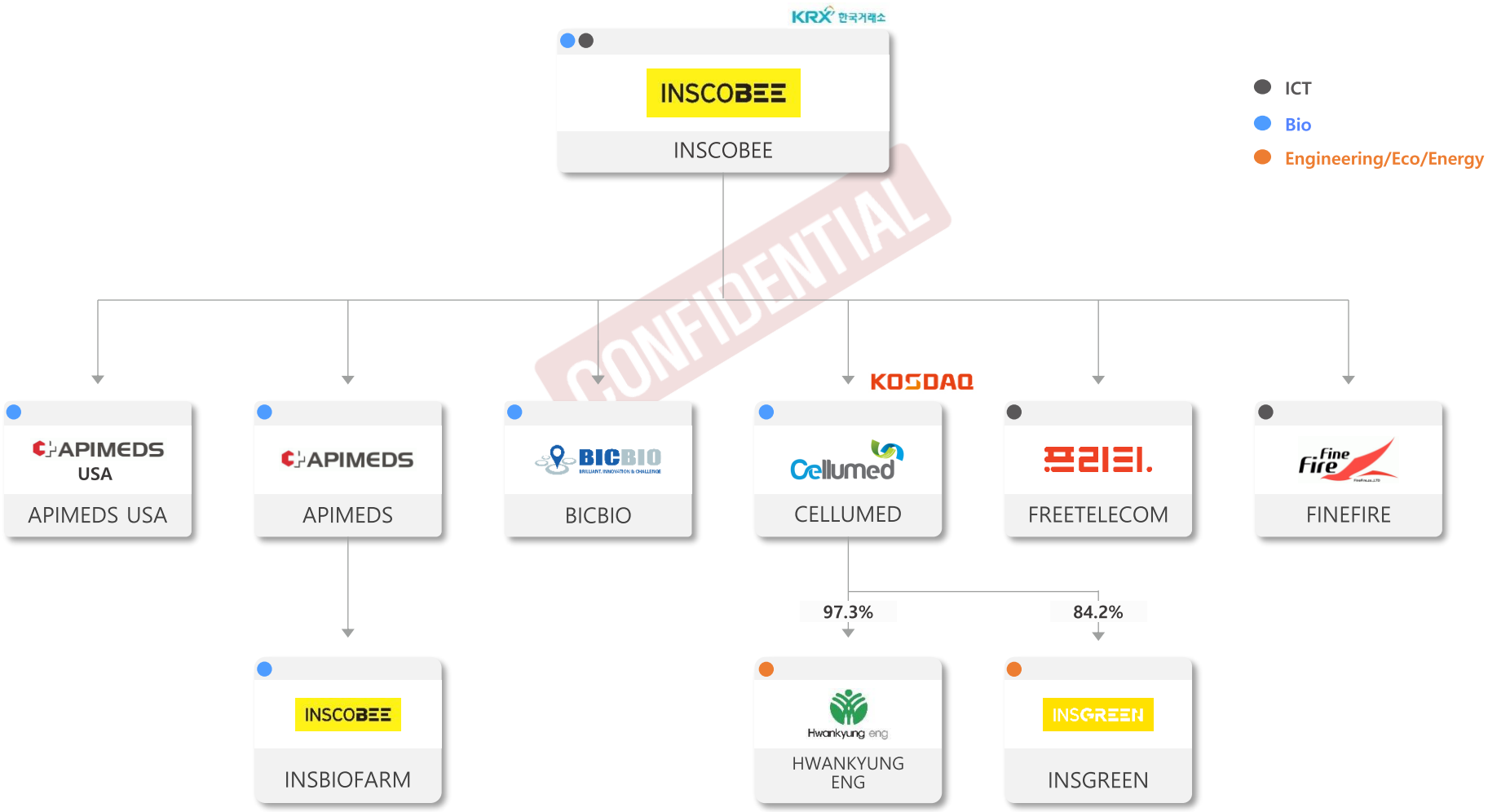
Introduction of Affiliates



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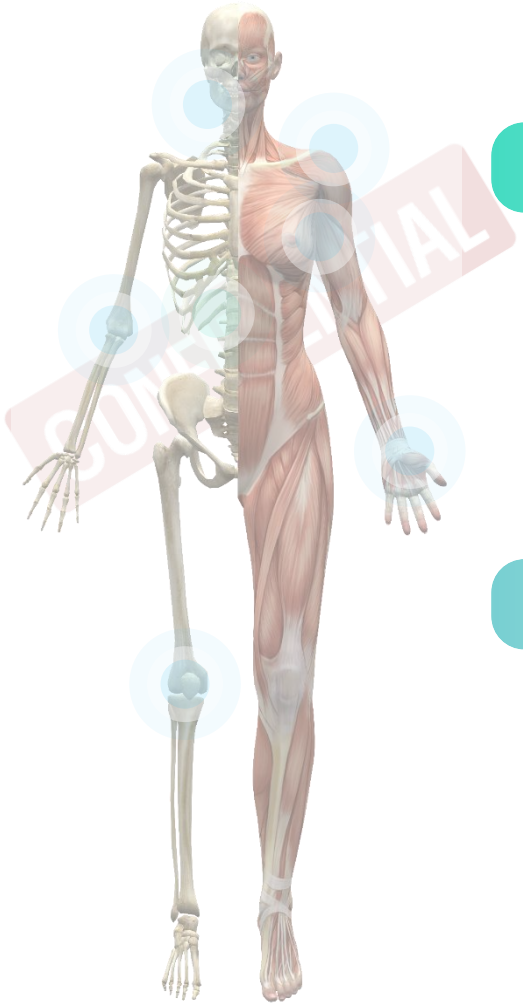
Business Area



Tissue Bank

Distribution and processing of human tissues such as bones, ligaments, and skin

* Human tissues are used in their original form or with minimal processing, and are mainly used to repair structural/functional defects.

RNA

GMP-grade RNA enzyme production / RNA therapeutics development

Medical Devices

Processing and manufacturing sales of allogeneic bone fillers / Sales of artificial knee joints

* Human tissue-based medical devices are products designed to enhance function or serve a specific therapeutic purpose through additional processing and treatment.

Distribution

ODF (Oral Dissolving Film) / Pharmaceuticals

Tissue Bank

Medical Devices

Production Process

Skin materials
→ Foundation Sterilization Q.C
→ **SKIN**

Bone materials
→ Cutting Dry Grinding Chemical Treatment
→ Lyophilization Sterilization Q.C
→ **Dental Powder**

→ Chemical Treatment PH Measurement Sterilization Viscosity Testing
→ **DBM**

Titanium alloy
→ TiN Coating
→ **TKR**

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Product Line

[CELLUDERM Series]

HD IMPLANT

ULTRA

POWER

FILL

GEN

[Dental Powder]

* ODM

Rafugen DBM Gel Pro

Bio BMP2

Luminus Flex

Chapter 2. Investment Highlights

Investment Highlights

Established GMP production facility

GMP

- GMP (GTP)-certified medical device production facility in Seoul
- Tissue bank import and manufacturing capabilities (skin, bone)
- Annual production capacity: Approximately 120,000 ▲ CC (based on DBM)
- Production facility-based development of plastic surgery and skin-related products (e.g., SKIN Booster)

RNA-based therapeutics and protein development

R&D

- Research and development of growth factors, siRNA, and mRNA therapeutics
- Joint development of therapeutics with renowned domestic researchers
  - Sogang University, Ewha Womans University / Vaccines, cancer therapeutics
- Possesses mRNA production enzyme technology
- Growth factor cell lines: BMP2, FGF7, TGF-B, etc.
- Growth factor-based business and commercialization references

Established a stable raw material and distribution network

Network

- Secured global raw material (bone, skin) suppliers: Solvita, Allosource
- Secured nationwide network of major distributors
- Major suppliers
  - Seoul National University Hospital, Yonsei Severance Hospital, Seoul Samsung Hospital, and etc.

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Growth (mid to long-term) Strategy

- ✓ By strengthening medical device and pharmaceutical distribution lineup, building on its orthopedic and dental sales base.
- ✓ By promoting global sales of mRNA-producing enzymes (T7 RNA Polymerase, DNase I).

- ✓ Develop "SKIN Booster" based on a skin tissue bank, as well as other dermatological and cosmetic developments
- ✓ AI-based DBM booster for fracture treatment
- ✓ Develop therapeutics based on growth factor (BMP2, FGF7) technology
- ✓ Develop siRNA and mRNA-based therapeutics and LOA (Localization of RNA)
  - Joint research underway with Sogang University and Ewha Womans University
  - Indications: Pulmonary fibrosis, pancreatic cancer (PDAC) treatment, universal influenza vaccine
  - Post-clinical LO planned

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A Stable Network in Orthopedics and Dentistry

Strengthen medical device and pharmaceutical product lineup

Expand plastic surgery and skin care sectors

Treatment LO

Plastic Surgery Market Outlook

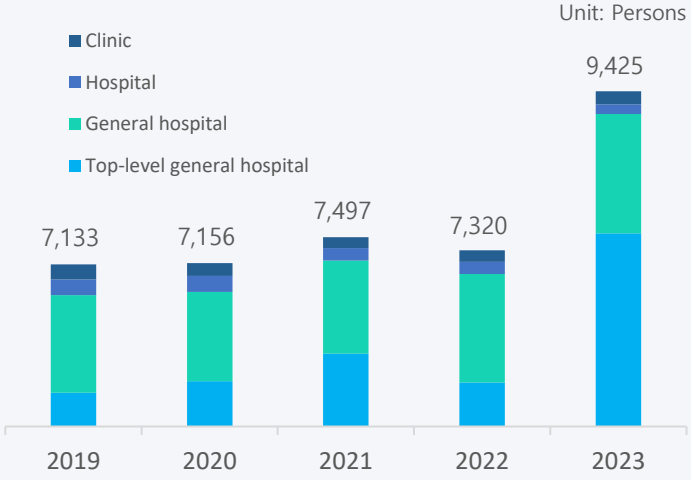
- Growing demand for plastic surgery and technological advancements are driving growth. The market is estimated to grow from approximately $1.5 billion in 2023 to approximately $4.6 billion in 2032, with a compound annual growth rate of approximately 13.2%.

Key growth factors

- The rise in breast cancer patients is driving an increase in breast reconstruction procedures.
- Expand dermatology and plastic surgery market with the development of aesthetic medical devices
- Develop tissue banks based skin boosters

More Clean, More Accurate, More Diverse

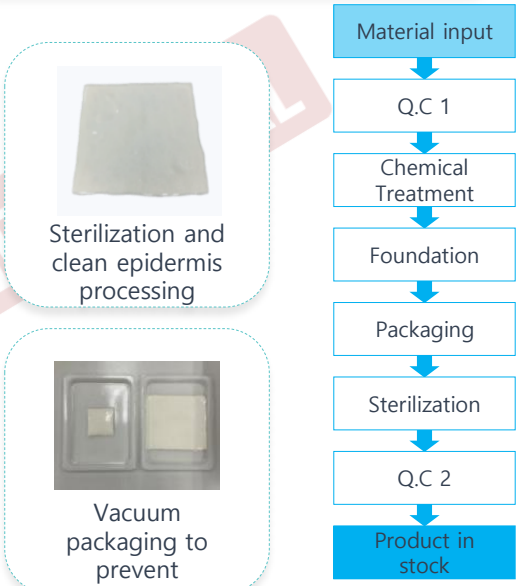
Number of breast reconstruction surgeries performed in Korea



Reconstruction rate among mastectomy patients

50%

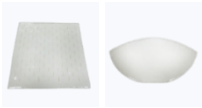
SKIN production process diagram



Sterilization and clean epidermis processing

3D scanners ensure accurate size and thickness control

Vacuum packaging to prevent deformation



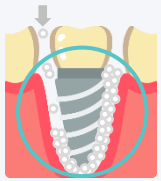
Customized and available in various shapes

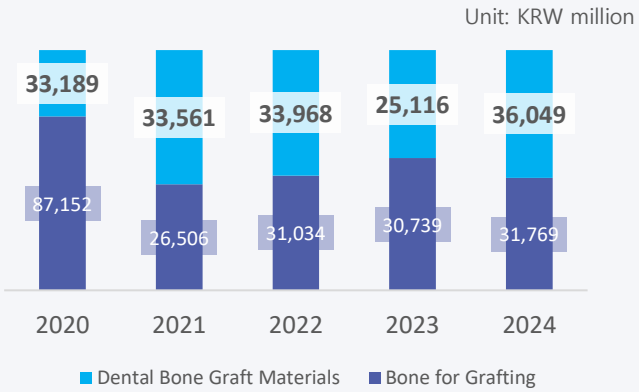
Implant, Inject Type
Expanded indications

High-Quality Bone Graft Material Satisfying both Clinicians and Patients

Korean dental bone graft market

* Dental bone graft materials are used during implant procedures to secure a sufficient amount of bone tissue or to increase bond strength with surrounding tissue





▶ With the expansion of implant procedures with aging, demand for dental bone graft materials used in procedures is expected to increase steadily

Source : Ministry of Food and Drug Safety (MFDS)

Dental powder manufacturing process

Provides more volume compared to competing products (e.g. 0.3cc)

RENEW MEDICAL
DENALL

Supply to famous customers and dentists nationwide

Raw material input
↓
Q.C 1
↓
Cutting
↓
Chemical treatment ①
↓
Drying and milling
↓
Chemical treatment ②
↓
Freeze drying
↓
Packaging
↓
Sterilization
↓
Q.C 2
↓
Finished goods

8 : 2

Cortical bone Cancellous bone

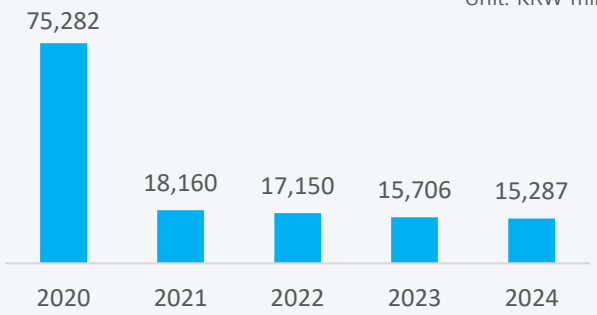
Optimal ratio for filling bones and making bones

It is easy to use as it is a syringe type

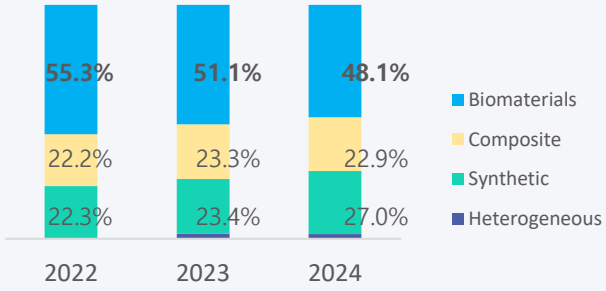
Higher DBM content

Korean Bone Graft Biomaterials Market

Unit: KRW million



Ratio of Material Use (3 years)



Source : Ministry of Food and Drug Safety (MFDS)

DBM _ Rafugen DBM

- Allograft bone graft material designed to promote superior bone formation
- Greater DBM content ratio compared to competing products
- US FDA 510k certified FDA

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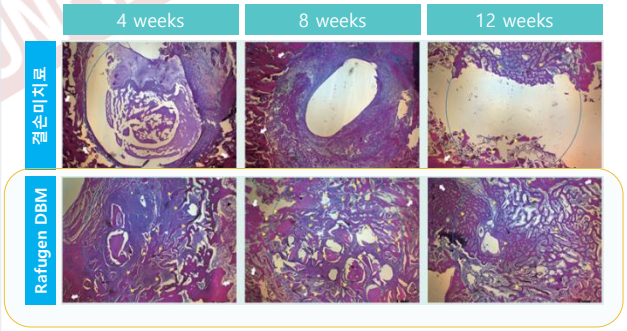
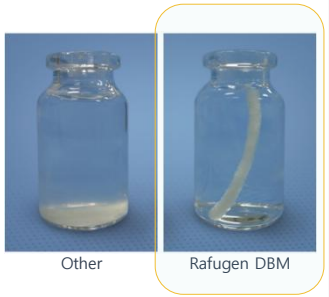
Product feature tests



- Rafugen DBM was implanted into rabbit femoral bone defects; after 12 weeks, bone formation is clearly shown compared to untreated defect sites



- Shape retention of Rafugen DBM confirmed after incubation in distilled water for 24 hours

!

- Excellent bone induction using high DBM content
- Strong viscosity prevents DBM loss and derives excellent moldability

Stable Supply and Profit Management with Increased Production Volume

Manufacturing Process

Raw Material Input → Q.C 1 → Cutting → Chemical treatment ① → Drying and milling → Chemical treatment ② → Chemical treatment ③ → pH measurement and drying →

Packaging → Sterilization → Q.C 2 → Gel filling and Packaging → Freeze → Sterilization → Viscosity test → Finished goods

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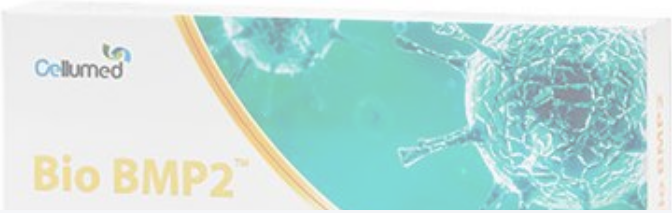
DBM production plan

1st Construction : Installment of new equipment

2nd Construction : Facility construction

Safe rhBMP2 Protein Produced Using Mammalian Cell Culture



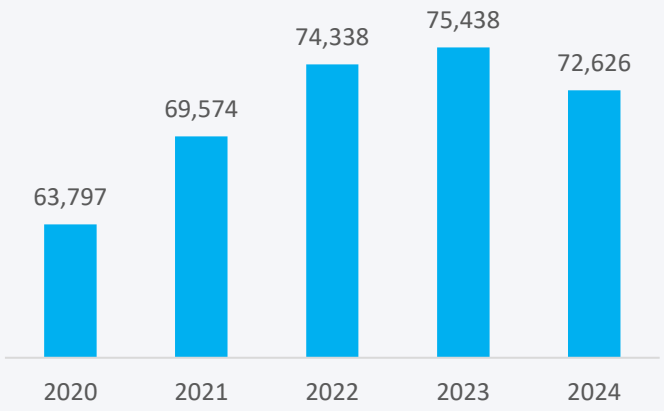
- Bone graft material containing osteoinductive proteins in DBM
- Korea's first CHO cell-derived glycoprotein approach for enhanced safety

Mammalian cells (CHO)		Escherichia coli (E.coli)
• Maintain a three-dimensional structure with human-like proteins • Enables complex protein folding and saccharification • Can be modified after translation such as saccharification, acetylation, phosphorylation, and etc.	Protein structure	• Fail to form complex three-dimensional structures • Frequent protein folding errors and no glycosylation • Post-translational modification required for human proteins is not possible
5 to 25 times higher activity compared to E. coli	Protein activity	Reduced protein biological activity
Human-like structure with low chance of developing an immune response	Biocompatibility	May potentially trigger an immune response

Lightweight, long-lasting Total Knee Replacement using TiN-coated Titanium

Korean total knee replacement market

Unit: KRW million



▶ As human life expectancy increases, the demand for artificial knee joints is expected to continue to increase

Source : Ministry of Food and Drug Safety (MFDS)

TKR _ Luminus Flex Total Knee Replacement

- Lightweight and improved wear resistance via TiN-coated titanium
- The only product using titanium for both femoral and tibial components
- Designed for Asian anatomy and sedentary lifestyles with 150° flexion

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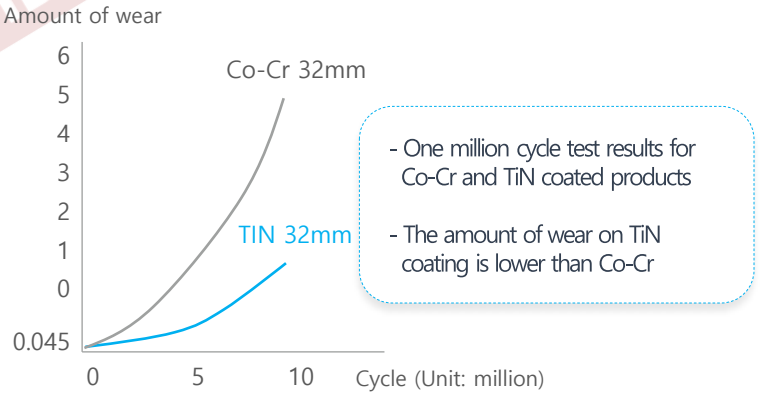
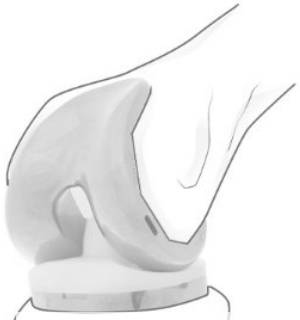
Wear resistance of TiN coating



! Its lightweight and high biocompatibility using titanium
TiN coating can improve wear resistance and provide longer durability which renders patients with positive influence on daily activities

TKR

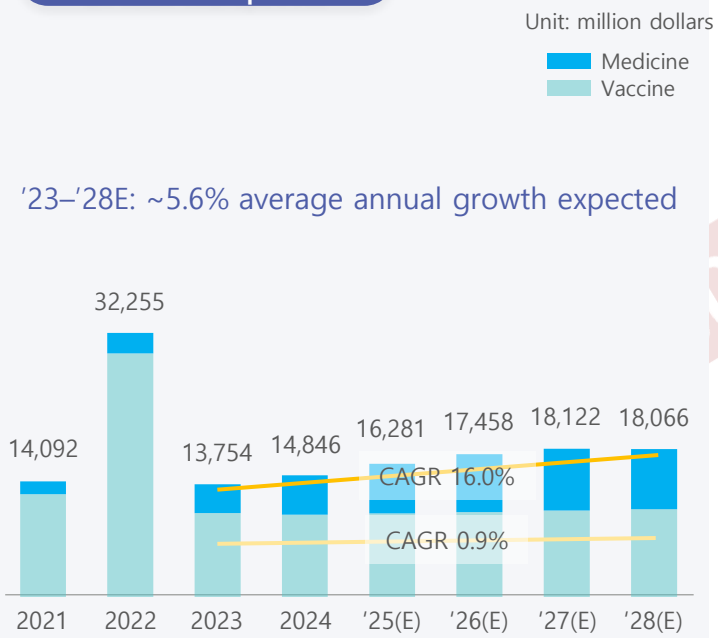
* TiN (titanium nitride) coating: A ceramic-like thin film coating formed by combining titanium (Ti) and nitrogen (N), which improves the wear resistance and biocompatibility of metal surfaces



TiN coated Ti-6Al-4V		Co-Cr
Titanium TiN coating		**Cobalt chrome**
Stiff (Surface stiffness is enhanced using TiN coating technology)	**Stiffness**	Very stiff (Capable of supporting high loads)
Light (4.4~4.5 g/cm³)	**Weight**	Heavy (8.3~9.2 g/cm³)
Very low coefficient of friction (0.02~0.05)	**Wear resistance**	Medium coefficient of friction (0.05~0.08)
Very superior (Corrosion prevention using TiN layer)	**Corrosion resistance**	superior
Prevention of Metal Allergy Bone fusion on titanium	**Osteosynthesis** **Osteointegration**	Potential allergic reaction No bone fusion

RNA Enzyme Performance Evaluation Completed : Entering Commercialization Stage for GMP-grade Enzymes

RNA Global Market Based on RNA Therapeutics

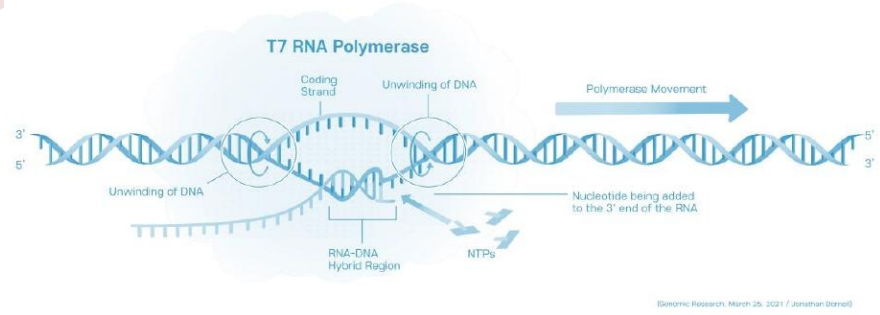


Source:Markets & Markets (2023)

RNA Enzyme

- RNA enzymes are biological catalysts involved in synthesis, processing, and degradation of molecules. Demand is increasing with the rapid growth of mRNA vaccines and RNA-based therapeutics.
- In partnership with Germany-based RHB, maintain in-house cell line banking, and supply enzymes for drug development and manufacturing

T7 RNA Polymerase



T7 RNA polymerase used in RNA production can be replaced with our product.

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RNA Therapeutics Development for Intractable Diseases

RNA Therapeutics

- RNA-LNP (lipid nanoparticle) therapeutics enable efficient intracellular delivery—reducing side effects and improving efficacy—by attaching target-binding ligands to the LNP surface
- Co-development with Sogang University Research & Business Foundation and Ewha University Industry Collaboration Foundation

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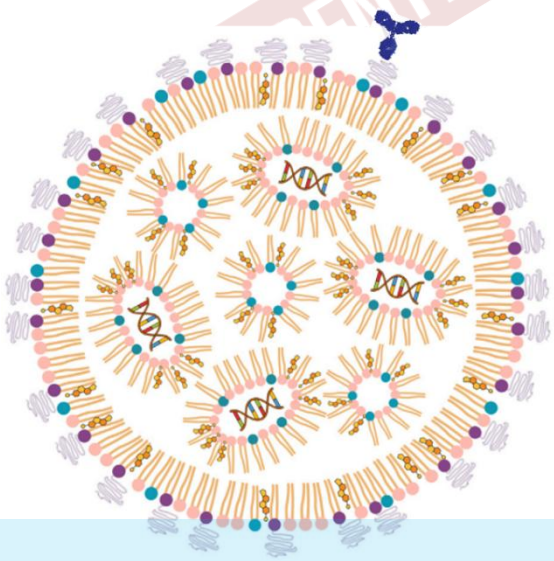
RNA-LNP

Target diseases

- IPF
- PDAC
- Ovarian cancer



R&D for RNA-LNP

- Treg cell
- siRNA
- Active targeting

! Idiopathic Pulmonary Fibrosis (IPF) / Pancreatic Ductal Adenocarcinoma (PDAC)

Development of 'DBM Booster' for Fracture Treatment

BMP2 Metabolic control fracture treatment

- Using AI (AlphaFold 3) to design functional peptide therapeutics that modulate BMP2 metabolism—reducing BMP2 degradation and increasing its plasma half-life—to support fracture healing
- Potential to reduce BMP2 dose and mitigate adverse effects from high initial burst release
- Sustained-release formulation expected to provide consistent and durable efficacy

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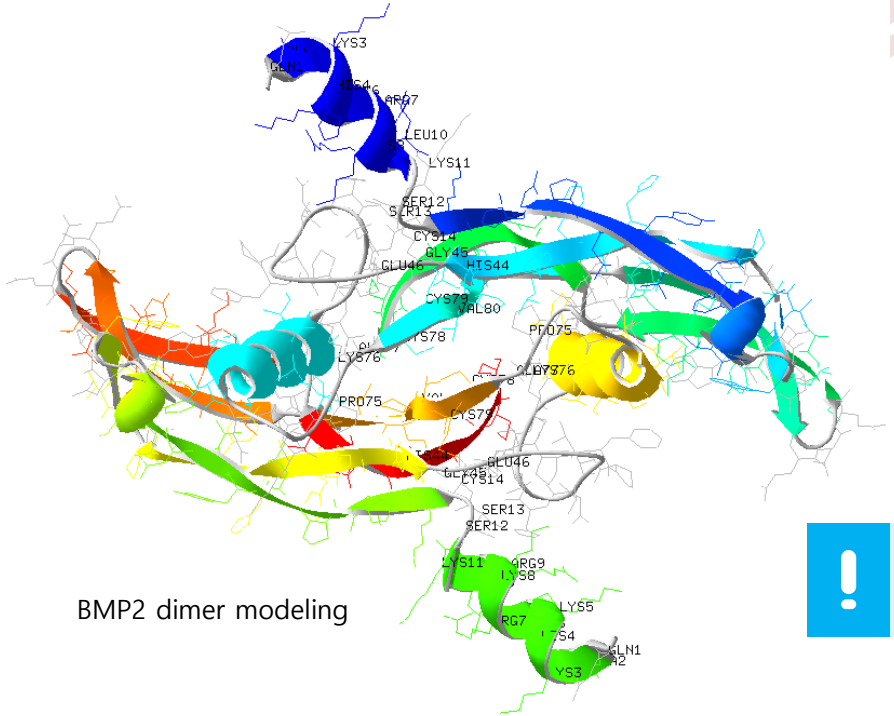
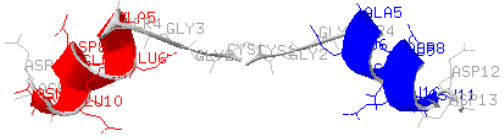


BMP2 dimer modeling



De novo peptide design

Development of New Peptide Drugs for Fracture Treatment Using Artificial Intelligence

Other Distributions

Diversification and Expansion of Business

ODF _ Film formulation that quickly melts on your tongue without having to swallow water

- Absorbed directly into the body through the blood vessels of the eastern mucosa.
- High bioavailability with minimal impact from digestive enzymes
- Easily stored without water, making it convenient to carry and store.

Apitoxin _ Dried bee toxin (1 milligram)

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- Korea's first natural product drug
- Dried honeybee venom extracted, purified, and quantified using electroconvulsive technology
- Relieves pain caused by joint disease

Affiliate

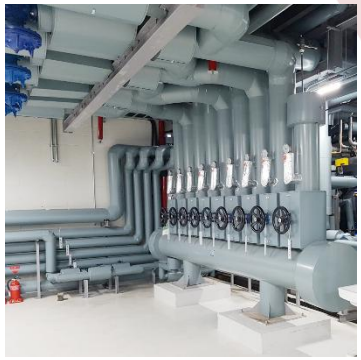
Company overview

Hwankyung eng

Possesses excellent technical skills in the field of mechanical equipment and is designated as an excellent partner by a number of acknowledged construction partner companies

- **Date of establishment :** March 1, 1985
- **CEO :** Yoo Chang-soo
- **Business :** Mechanical equipment construction, automatic control unit construction, Fireworks and Gas Facilities Corporation
- **Website :** www.hk-eng.com

Business



Mechanical equipment

Registrated Date: Dec. 18th 1982
Registration No: 82-12-245
Construction capacity: 69,848,000,000 KRW
National Contractor Ranking: 52nd / 7411 (2020)

Fire fighting equipment

Registrated Date: Sept. 11th 1996
Registration No: 1985-2
Construction capacity: 43,166,500,000 KRW
National Contractor Ranking: 49th / 5561 (2020)

Maintenance of facilities

Registrated Date: Feb. 4th 2013
Registration No: 13-29-01
Construction capacity: 8,335,808 KRW
National Contractor Ranking: 335th / 6344 (2019)

Affiliate

Company overview

INSGREEN

INSGREEN

Possesses excellent technical expertise and extensive experience in executing construction projects within the environmental plant and construction sectors

- **Date of establishment :** March 24, 1988
- **CEO :** Jin-Yeol Kim
- **Business :** Mechanical equipment / environmental specialist / civil engineering and construction
- **Website :** www.insgreen.co.kr

Business

Development

Logistics/residential, commercial/resource

Construction

Architecture/wood/ earthwork/electricity/painting

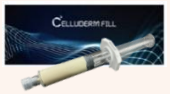
Environment

Water treatment/waste resource/consignment operation

Appendix

CELLUDERM Series

- A set of products that removes only cellular antigens responsible for immune responses while preserving proteins/components and the 3D structure of the dermal layer
- With low antigenicity, pro-angiogenic capability, and stability, it is used to reconstruct skin defects caused by burns, trauma, or other diseases

Unit	HD IMPLANT	ULTRA	POWER	FILL	GEN
					
Type	Hydrated Type	Freeze Dried Type		Syringe Type	
Ingredient	Allogeneic dermis	Allogeneic dermis		Fine dermal powder(30wt%) + sterile purified water	Fine dermal powder(5wt%) + sterile purified water
Characteristics	• No additional hydration and defrosting required(Ready to Use) • Rapid initial proliferation rate of cells	• Excellent tensile strength and elasticity • Stored at room temperature		• Direct injection into damaged soft tissue areas • Can be used regardless of the shape of the defect • Also known as joint injection and collagen injection, which can be used for simple surgical procedures. Patients may resume daily life immediately after such procedure	
Indications (Department)	• **Breast Reconstruction (Plastic Surgery)** • **Bedsores, diabetic foot ulcers (orthopedics)** • **Burns (General Surgery)**	• **Breast Reconstruction (Plastic Surgery)** • **Rotator Cuff Tear (Orthopedics)** • **Burns (General Surgery)** • **Genital Reconstructive Surgery (Urology)**		• **Breast Reconstruction (Breast Surgery)** • **Rhinoplasty (Plastic Surgery)** • **Soft Tissue Reconstruction (Orthopedics)** • **Genital Reconstructive Surgery (Urology)**	• **Soft Tissue Reconstruction (Orthopedics)** • **Soft Tissue Reconstruction (Neurosurgery)** • **Vaginoplasty (Obstetrics and Gynecology)**

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Consolidated Financial Statements

Consolidated Statement Of Financial Position

Unit: KRW million

Type	2025 3Q	2024	2023	2022
Current assets	55,495	52,293	50,057	65,554
Non-current assets	16,127	17,931	16,414	27,702
Non-current assets	**71,623**	**70,225**	**66,471**	**93,256**
Current liabilities	52,985	58,420	39,463	33,898
Non-current liabilities	2,116	2,744	3,990	4,228
Total liabilities	**55,101**	**61,164**	**43,453**	**38,127**
Equity attributable to owners of parent	16,308	8,777	22,461	53,945
Issued capital	27,479	25,758	21,402	20,888
Capital surplus	137,012	133,743	120,176	114,294
Elements of other stockholder's equity	(11,014)	(9,595)	(9,640)	(9,050)
Retained earnings	(137,168)	(141,129)	(109,477)	(72,188)
Non-controlling interests	213	284	556	1,185
Total equity	**16,522**	**9,061**	**23,018**	**55,129**

Consolidated Statement Of Comprehensive Income

Unit: KRW million

Type	2025 3Q	2024	2023	2022
Revenue	**111,915**	**129,140**	**95,553**	**122,085**
Cost of sales	107,775	119,650	89,921	108,454
Gross profit	**4,140**	**9,489**	**5,632**	**13,631**
Selling general administrative expenses	8,619	16,321	16,090	16,359
Operating income	**(4,479)**	**(6,832)**	**(10,458)**	**(2,728)**
Other gains	10,411	3,252	5,243	7,766
Other losses	1,741	27,024	33,366	9,838
Profit before tax	**4,084**	**(32,085)**	**(38,581)**	**(4,800)**
Tax expense	2	(267)	(654)	384
Profit	**4,802**	**(31,817)**	**(37,945)**	**(5,387)**

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Separate Financial Statements

Statement of Financial Position

Unit: KRW million

Type	2025 3Q	2024	2023	2022
Current assets	14,674	13,511	22,043	27,995
Non-current assets	33,016	33,081	33,138	44,340
Non-current assets	**47,690**	**46,592**	**55,181**	**72,334**
Current liabilities	23,137	30,946	23,443	18,167
Non-current liabilities	1,753	2,181	3,724	3,602
Total liabilities	**24,890**	**33,127**	**27,167**	**21,769**
Issued capital	27,479	25,758	21,402	20,888
Capital surplus	138,125	134,856	121,289	115,386
Elements of other stockholder's equity	(8,938)	(8,971)	(9,016)	(8,446)
Retained earnings	(133,866)	(138,177)	(105,661)	(77,263)
Non-controlling interests	**22,800**	**13,465**	**28,014**	**50,566**

Statement Of Comprehensive Income

Unit: KRW million

Type	2025 3Q	2024	2023	2022
Revenue	**9,149**	**11,167**	**15,146**	**27,360**
Cost of sales	6,696	9,198	11,332	21,634
Gross profit	**2,454**	**1,968**	**3,814**	**5,726**
Selling general administrative expenses	5,821	9,485	10,599	9,326
Operating income	**(3,367)**	**(7,516)**	**(6,785)**	**(3,600)**
Other gains	9,114	2,405	5,024	3,182
Other losses	1,436	27,324	26,722	6,285
Profit before tax	**4,311**	**(32,436)**	**(28,483)**	**(6,703)**
Tax expense	-	-	-	-
Profit	**4,311**	**(32,436)**	**(28,483)**	**(6,703)**

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